

Lóránt Nagy, PhD

Budapest, Hungary

lorantnagy@renyi.hu

Professional Experience

Artificial Intelligence National Laboratory

Research Assistant

2024

Budapest, Hungary

- Worked with ALTEO on day-ahead electricity-price forecasting.
- Developed and evaluated machine-learning regression models and assessed their reliability.
- Communicated modelling choices, limitations, and findings to academic and industry collaborators.

Eötvös Loránd University, AI Research Group

Research Assistant

2023–Present

Budapest, Hungary

- Conduct research on machine-learning methods for stochastic processes and long-memory time series.
- Conducted an interpretability study of neural-network-based Hurst estimation, providing evidence that the model relies primarily on spectral rather than geometric information.
- Develop reinforcement-learning methods for trading under market frictions, comparing learned performance with theoretical asymptotic behaviour.

Alfréd Rényi Institute of Mathematics

Research Assistant

2019–Present

Budapest, Hungary

- More recent work focuses on generative AI and deep learning, particularly diffusion-based generative models.
- Developed and presented a framework for studying superlinearly mean-reverting diffusions as alternative forward processes in generative modelling, with ongoing numerical experiments comparing them with classical Ornstein–Uhlenbeck dynamics.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Morgan Stanley & Co. LLC

Interest Rate Modelling Intern

2017–2018

Budapest, Hungary

- Applied quantitative methods to problems in interest-rate modelling.
- Gained experience with mathematical modelling and numerical analysis in an industry setting, as well as communicating technical work within a professional team.

Education

Central European University

PhD in Mathematics

2018–2022

Budapest, Hungary

Budapest University of Technology and Economics

MSc in Mathematics

2015–2018

Budapest, Hungary

Technical Skills

Programming and development tools: Python, PyTorch, Git, Docker, GitHub Actions, Wolfram Mathematica, LLM-assisted development workflows

Quantitative methods: stochastic modelling, simulation, optimization, statistical analysis, and mathematical finance

Cloud and deployment: Azure Static Web Apps, Azure Container Apps; basic CI/CD workflows and containerized application deployment

Machine learning tools and workflows: Weights & Biases, TensorBoard, reproducible experiment workflows, model comparison, and evaluation

Machine learning and data analysis: deep learning, neural networks, diffusion-based generative modelling, regression, and time-series forecasting

Publications

- Rásonyi, M. and Nagy, L. (2021). *Optimal long-term investment in illiquid markets when prices have negative memory*. Electronic Communications in Probability, 26.
- Guasoni, P., Nagy, L., and Rásonyi, M. (2021). *Young, timid, and risk takers*. Mathematical Finance, 31, 1332–1356.
- Boros, D., Csanády, B., Ivkovic, I., Nagy, L., Lukács, A., and Márkus, L. (2024). *Deep learning the Hurst parameter of linear fractional processes and assessing its reliability*. Quality and Reliability Engineering International, 40, 4228–4246.
- Csanády, B., Nagy, L., Boros, D., Ivkovic, I., Kovács, D., Tóth-Lakits, D., Márkus, L., and Lukács, A. (2024). *Parameter estimation of long-memory stochastic processes with deep neural networks*. Proceedings of ECAI 2024.
- Rásonyi, M. and Nagy, L. (2025). *Optimal investment with transient price impact and stochastic spread*. Preprint, arXiv:2511.12093.
- Rásonyi, M. and Nagy, L. *On arbitrage with linear and superlinear friction*. Manuscript in preparation.

Conference and Seminar Presentations

- 2026** *On the utility problem in a market where price impact is transient*
XIII Bachelier World Congress, Bologna
- 2025** *Non-convex optimal investment with transient price impact*
Vienna Congress on Mathematical Finance, TU Wien
- 2024** *Optimal execution in superlinearly mean-reverting environments*
AMK Seminar, Bolyai Institute, University of Szeged
- 2023** *Risk-averse trading*
ELTE–Rényi AI Workshop, Eötvös Loránd University
- 2023** *Random walks in random environments*
Probability Seminar, Alfréd Rényi Institute of Mathematics
- 2022** *Young, Timid, and Risk Takers*
Financial Mathematics Seminar, University of Michigan
- 2021** *On sub-Gaussian estimators of the mean*
Probability Seminar, Alfréd Rényi Institute of Mathematics
- 2020** *Investment in mean-reverting markets*
PhD Seminar, Central European University
- 2020** *How to invest in mean-reverting markets?*
Probability Seminar, Alfréd Rényi Institute of Mathematics

Additional Experience and Distinctions

- 2025** Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office.
- 2022** Award for Advanced Doctoral Students.
- 2019** Instructor, *Fundamentals of Stochastic Analysis*, Central European University.
- 2016** Second Place, TDK competition, *Examples in the theory of continuous trading with friction*.

Things I Build Outside Work

- **Slow-paced science-film project for kids on YouTube (@CrowburnFilms)**: Experiment with generative-image and generative-video workflows, including running open-source video-generation models on Runpod GPU infrastructure using ComfyUI and WAN 2.2.
- **PageIt** (pageit.me): A small web application using Azure Static Web Apps and Azure Container Apps, with Docker and GitHub Actions for automated frontend deployment and backend image build and publishing.